MIL-R-36334A (DSA-DH)
20 April 1973
SUPERSEDING
MIL-R-36334 (DM)
21 November 1964

MILITARY SPECIFICATION

REGULATOR, SUCTION, SURGICAL SUCTION APPARATUS

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of regulatory apparatus, suitable for controlling suction in the drainage of postoperative thoracic surgery patients.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

ZZ-S-751	Stopper, Bottle, Rubber.
~~MMM~~-A-250	Adhesive, Water Resistant (For Sealing Waterproof Paper).
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

FSC 6515

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:25Z
Check the source to verify that this is the current version before use.

STANDARDS

Federal

Fed. Std. No. 66	Metals; Chemical Composition and Hardenability.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Shall be of the highest quality throughout, and suitable in all respects for the purpose intended. Gages specified are a minimum; commercial tolerances shall apply.

* 3.2 Construction. Construction of the regulator shall be as shown on Figure 1 and as hereinafter specified. The regulator shall be comprised of the following components. Tolerance on length dimensions shall be ± 1/16 inch.

3.2.1 Tubes. Inlet, outlet, and suction, tubes shall be of corrosion resisting steel conforming to 302 of Fed. Std. 66, 0.032 inch thick, No. 20 Brown and Sharp gage (B&S).

3.2.2 Cover plate. Shall be of corrosion resisting steel conforming to 302 of Fed. Std. 66, 0.040 inch thick No. 18 B and S gage.

3.2.3 Retainers. Shall be of rubber, and capable of sterilization.

3.2.4 Washers. Shall be of a suitable felt.

3.2.5 Stoppers. Shall conform to Type B, grade 1 of ZZ-S-751.

3.3 Resistance to corrosion. All metal component parts shall be corrosion resistant.

3.4 Assembly. The assembly of the suction regulator shall conform with the figure 1. The lubricated suction adjustment tube shall be stationary when no pressure is applied, and shall move freely when pressure is applied. All parts shall be interchangeable, and shall align and fit properly. The regulator shall be furnished without lubricant.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:25Z
Check the source to verify that this is the current version before use.

3.5 Performance. The suction regulator shall be suitable for controlling suction from the chest of postoperative thoracic patients. The regulator shall not leak when tested as specified in 4.4.1.

* 3.6 Finish. Corrosion-resisting steel shall have a satin finish.

* 3.7 Identification marking. Each suction regulator shall be permanently and legibly marked with the manufacturer's name or registered trademark.

3.8 Workmanship. Workmanship shall be first class throughout. Regulators shall be free from defects which detract from their appearance of may impair their serviceability. The lubricated suction adjustment tube shall be stationary when no pressure is applied, and shall move freely when pressure is applied.

3.9 Instructions. Comprehensive instructions shall be furnished with each regulator.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, and auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:25Z
Check the source to verify that this is the current version before use.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the tubes, cover plate, retainers, washers and stoppers may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the Federal Stock Number, the Item Identification, the name of the component material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and Table I.

Table I. Sampling for Examination

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	S-2	2.5
Minor defects	S-4	3.5
For dimensional examination	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, inspection level S-1.

Table II. Sampling for Tests.

Component (unit of product)	Characteristic	Requirements	Test procedure	AQL
End item	Leakage	3.5	4.4.1	*

* Acceptance number shall be zero.

4.3 Examination. The regulators shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects, as shown in Table III.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:25Z
Check the source to verify that this is the current version before use.

Table III. Categories and defects*

Categories	Defects
Major	
101	Regulator not free of sharp edges.
102	Holes in cover plate and stopper not aligned.
103	Holes in stopper not clean cut.
104	Rubber surfaces not free of stickiness.
Minor	
201	Metal surfaces not free of scratches, pits, and grind marks.
202	Rubber surfaces not free of nicks.

*Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The regulators shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Leakage. When the regulator is placed tightly into a suitably sized bottle there shall be no discernable leakage of air into the bottle, when tested as follows:

(a) Insert stopper tightly into a suitable bottle.

(b) Attach a tight fitting rubber tube to outlet tube and the other end to a vacuum indicator.

(c) Attach one end of a tight fitting rubber tube to the inlet tube and the other end to a source of vacuum. A pinch cock on this rubber tube shall be used for shutting off the vacuum at the desired time.

(d) Seal the inlet to the adjustment tube and turn on the vacuum.

(e) When 15 inches of mercury vacuum is reached, turn off the vacuum and apply the pinch cock. The vacuum gage should show no appreciable decrease in value for five minutes.

Source: https://assist.dla.mil -- Downloaded: 20261O-21T08:25Z
Check the source to verify that this is the current version before use.

* 5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One regulator, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in the following table shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	6 units	72 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class domestic. Box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to the contents. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closure of bottom flap with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:25Z
Check the source to verify that this is the current version before use.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.2. Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Thoracic surgery in military medical facilities.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:25Z
Check the source to verify that this is the current version before use.

6.3 This specification covers the following item appearing in the Federal Supply Catalog:

Stock Number	Item Identification
6515-00-299-8339	Regulator, Suction, Surgical Apparatus

6.4 This specification does not include all types, classes, grades and sizes, of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types which are normally procured to meet Federal requirements.

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1547

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document draft circulation should be based on the information in the current DODISS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:25Z
Check the source to verify that this is the current version before use.

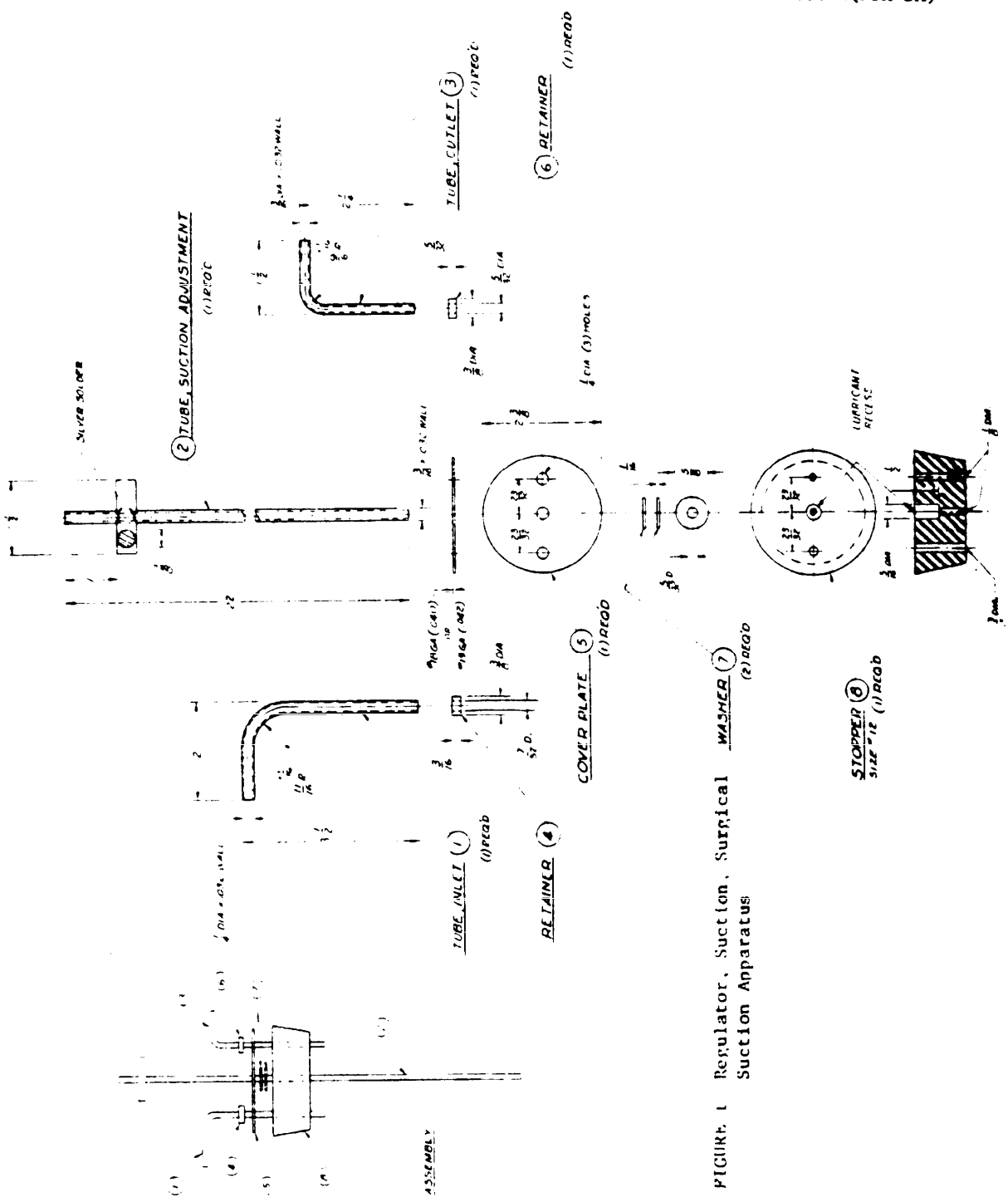


FIGURE 1. Regulator, Suction, Surgical Suction Apparatus

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:25Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:25Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers' vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-R-36334A (DSA-DM)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

YES NO (If "Yes", in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:25Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE PERSONNEL SUPPORT CENTER
2800 South 20th Street
Philadelphia, Pa. 19101

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY
DoD - 304

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATI
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:25Z
Check the source to verify that this is the current version before use.